Stabilizing Cortical Wetware by Natural Algorithm Optimization

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Abstract
Briefly summarize the research question, methods, principal results, and conclusions.

1 Introduction

Cortical wetware is the use of human neurons, typically derived from iPSC stem cells, due to recent developments in both ease and efficiency in the chemical method [1].

2 Related Work

3 Methods

3.1 Data, preprocessing and in-silico model definition

The entire analysis is performed using a 1000-neuron connected cell-network within an extracellular matrix, all simulated using NetPyNE and NEURON [2] [3]. The set-up for the cell was based upon the need to develop close-to-reality exchange of ions and GABA receptor dynamics which induce the unstable activity we wish to resolve.

Seizure models were deried from single-cell models, which are then applied to the wider system. In order to systematize the single-cell models, NetPyNE was parameterized to adopt <fill in here>.

3.2 Analysis

The "accuracy" or performance of the model was evaluated based on specific parameters.

3.3 Algorithms

3.3.1 Neural Nets

3.3.2 Ant Colony Optimization

3.3.3 Bee Colony Optimization

3.3.4 Artificial Immune Systems

3.3.5 Firefly Algorithms

3.3.6 Particle Swarm Optimization & Genetic Algorithms

4 Results

Present the findings without interpretation. Use tables and figures as needed (e.g., Figure 1).

Table 1: Example results table.

Method	Metric 1	Metric 2
Baseline	–	–
Proposed	–	–

Figure 1: Example figure caption.

5 Discussion

Interpret the results, compare them with prior work, and discuss implications.

6 Limitations

Describe assumptions, sources of uncertainty, and limitations of the study.

7 Conclusion

Summarize the main findings and identify promising directions for future work.

Acknowledgments

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AI Declaration

Some of the codebase was developed and debugged via the use of Claude Opus 5.5 at high effort, a LLM from Anthropic PBC, San Francisco. Some literature search was conducted by Claude Sonnet 5.5 at high effort. All writing and conception was done by hand.

References

[1] Zhang, Yingsha et al. Rapid single-step induction of functional neurons from human pluripotent stem cells. Neuron vol. 78,5 (2013): 785-98. doi:10.1016/j.neuron.2013.05.029.

[2] Hines, Michael, Thomas Carnevale, and Robert A. McDougal. "NEURON Simulation Environment." Encyclopedia of Computational Neuroscience, edited by Dieter Jaeger and Ranu Jung, Springer, 2022, doi: 10.1007/978-1-0716-1006-0_795.

[3] Dura-Bernal, Salvador et al. NetPyNE, a tool for data-driven multiscale modeling of brain circuits. eLife vol. 8 e44494. 26 Apr. 2019, doi:10.7554/eLife.44494.